AUBAID AHMED SAIYED

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Summary

Pre-final year IT student building full-stack applications with Python, JavaScript, Node.js, and React.js, grounded in Data Structures Algorithms and System Design. Hands-on experience with RESTful API architecture, database management (MongoDB, MySQL), caching strategies, and distributed backend design.

Experience

Social Summer of Code'26 May 2026 – June 2026
Contributor *Remote*

- Eliminated file-not-found failures by **building a centralized ingestion layer** with job-scoped deterministic storage and streaming file uploads, ensuring reliable, consistent file access across all pipeline stages.
- Connected an existing 6-stage **data processing pipeline** (OCR -> Vision -> Source Lookup -> Tagging -> Storage -> Indexing) through a **background job worker queue**, enabling fully **asynchronous, event-driven execution** of incoming uploads.
- **Improved system fault tolerance and observability** by adding per-stage error isolation, job lifecycle tracking, and detailed stage-level performance logging, and exposed **REST API** endpoints for real-time job status monitoring and debugging.

Hacktoberfest'25 Oct 2025 – Oct 2025
Contributor *Remote*

- Contributed to **open-source frontend development** by **debugging asynchronous state management issues** in React components and improving overall UI responsiveness.
- Enhanced **UI/UX performance** and **responsive, cross-device compatibility** through targeted CSS and rendering optimizations.
- Built automation scripts for **structured data extraction** and explored **API rate-limiting** strategies for third-party integrations.

Projects

Vector AI | Tech Stack: MongoDB - Node.js - Express.js - React.js - Python | Github | Link

- Built a **retail demand intelligence platform** using **React.js, Express.js, MongoDB Atlas, FastAPI, Redis**, and **XGBoost** for **demand forecasting**, **machine learning-based analytics**, and **EOQ-based inventory optimization** via a **RESTful, microservices-oriented** architecture.
- Developed and benchmarked **XGBoost machine learning models** for time-series demand forecasting, achieving **17% MAPE** through structured backtesting and model evaluation.
- Reduced forecast **API latency** by **75%** (**792ms to 193ms**) through **Redis caching**, **MongoDB indexing**, and request-level **caching strategies** for performance optimization.
- Accelerated store-wide forecasting throughput by **87%** (from **6.85s to 0.9s**) through **parallel processing**, **concurrency**, and SKU-level **workload partitioning**.
- Improved **concurrent workload performance** by **78%**, reducing average latency from **756ms to 166ms** and p95 latency from 950ms to 209ms, through **database query optimization** and **request deduplication**.

StackLens | Tech Stack: JavaScript - React.js - Node.js - Express.js - Babel (AST) - Dagre | Github | Link

- Architected a **stateless static code analysis engine** using **Node.js** and **React.js** that parses **Abstract Syntax Trees (ASTs)** via **Babel** across 1,400+ file repositories, processing code entirely **in memory** to eliminate storage overhead and enable scalable concurrent analysis.
- Engineered an **O(1) adjacency list construction** using **hash-based indexing**, removing O(N^2) path-resolution patterns and reducing graph construction overhead to approximately 2ms in measured runs.
- Implemented **O(V+E) BFS/DFS graph traversal algorithms** with **Set-based operations** to compute dependency impact paths, detect **circular dependencies (cycles)**, and identify orphaned modules in large-scale codebases.

Safe Stay | Tech Stack: MongoDB - Node.js - Express.js - React.js | Github | Link

- Developed Safe Stay, a **MERN stack** (**MongoDB, Express.js, React.js, Node.js**) **emergency shelter management platform** connecting evacuees with temporary, verified housing during crises and disasters.
- Implemented secure **JWT-based authentication and authorization**, **RESTful API design**, and backend services using **Node.js** and **Express.js**.
- Designed a **scalable backend architecture** with custom middleware, file upload handling, centralized error handling, and production-ready deployment configuration on **Vercel**.
- Built a responsive frontend using **React.js, Vite**, and **Tailwind CSS** with **role-based access control (RBAC)** dashboards for hosts, evacuees, and admins.

Skills

Languages: Python, Java, JavaScript, HTML, CSS
Frameworks & Tools: React.js, Node.js, Express.js, FastAPI, Tailwind CSS, Vite, MongoDB, MySQL, Redis, XGBoost, Git/GitHub, Vercel
Concepts: RESTful API Design, System Design, Data Structures & Algorithms, OOP, Asynchronous Programming, Microservices, Caching, RBAC
Coursework: Database Management Systems, Computer Networks
Soft Skills: Communication, Teamwork, Leadership, Product Thinking

Education

Charotar University of Science and Technology, Anand, India 2024 – 2028
B.Tech in Information Technology

Kendriya Vidhyalaya Rajkot, Rajkot, India 2023 – 2024
Class 12, Grade: 93%